

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed and developed **AI behaviors** and **NPC scripting** for a high-mobility multiplayer action game, contributing to engaging enemy encounters.
- Created detailed design specifications and **gameplay documentation** for AI features, including technical implementation guidelines for behavior trees.
- Analyzed **gameplay data** and player behavioral metrics to identify opportunities for AI system improvements and enhance game balance.
- Collaborated with AI designers and gameplay engineers on **behavior tree development** and NPC creation within a distributed team environment.
- Prototyped and iterated on innovative AI concepts that enhanced player engagement, balancing technical constraints with creative vision.
- Supported the continuous refinement of combat encounters and overall gameplay pacing by analyzing player feedback and performance metrics.
- Containerized AI feature services using **Docker** and deployed them to a **Kubernetes** cluster (EKS), managing configuration and resource allocation.

SKILLS

AI/ML – AI Design, NPC Behavior, Behavior Trees, AI Systems, Gameplay Mechanics, Game Development Engines, Unreal Engine

Programming – Blueprint Scripting, Python, Java, SQL

Game Development – Multiplayer Game Design, Enemy Encounters, Player Experience, Game Balance, Combat Design

Data Analysis – Gameplay Data Analysis, Behavioral Metrics, Heatmaps, Player Feedback Analysis

Cloud/DevOps – Docker, Kubernetes, AWS (EC2/S3/Lambda/CloudWatch), Jenkins, Git

Methodologies – Agile, Scrum, Remote-First Collaboration, Technical Documentation

PROJECTS

AI Behavior Tree System for Action Game

- Designed and implemented a modular **AI behavior tree** system in **Unreal Engine Blueprints** to manage complex NPC decision-making for a fast-paced action game.
- Developed reusable AI modules for enemy pathfinding, combat tactics, and environmental interaction, reducing scripting time for new enemy types by **~30%**.
- Collaborated with senior designers to integrate AI systems, ensuring behaviors aligned with core gameplay mechanics and player expectations.

Gameplay Data Analysis Pipeline

- Engineered a **Python**-based data pipeline to ingest and process gameplay telemetry from multiplayer sessions, focusing on player movement and combat engagement metrics.
- Analyzed processed data using **pandas** and **NumPy** to identify patterns in player behavior and enemy encounter effectiveness, informing design adjustments.
- Generated reports on key performance indicators related to AI effectiveness and player engagement, presenting findings to the core game design team.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)